

Fourier Series - MUST KNOW formulae

AGH Electronics and Telecommunications

Kinga McNerney

AGH University of Science and Technology

May 11, 2024



Expanding 2π - Periodic Function into AGH a trigonometric Fourier series

If a 2π - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -\pi, \pi \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad (1)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad n \geq 1 \quad (2)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx, \quad n \geq 1 \quad (3)$$

where integrations are over any interval in x of the length 2π .



Expanding $2l$ - Periodic Function into AGH a trigonometric Fourier series

If a $2l$ - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -l, l \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l},$$

where

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \quad (4)$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx, \quad n \geq 1 \quad (5)$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx, \quad n \geq 1. \quad (6)$$

where integrations are over any interval in x of the length $2l$.



Expanding l - Periodic Function into AGH a trigonometric Fourier series

If a l - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -\frac{l}{2}, \frac{l}{2} \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{2\pi nx}{l} + b_n \sin \frac{2\pi nx}{l},$$

where

$$a_0 = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) dx \quad (7)$$

$$a_n = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) \cos \frac{2\pi nx}{l} dx, \quad n \geq 1 \quad (8)$$

$$b_n = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) \sin \frac{2\pi nx}{l} dx, \quad n \geq 1. \quad (9)$$

where integrations are over any interval in x of the length l .



The Dirichlet Conditions

Function f satisfies **Dirichlet Conditions** on $\langle a, b \rangle$ if

- 1 f is piecewise monotonic (ie. $\langle a, b \rangle$ can be divided into a finite number of subintervals on which f is monotonic);
- 2 f is continuous on $\langle a, b \rangle$ apart from at most a finite number of points, at which it has discontinuities of the first kind;
- 3 at each point $f(x_0) = \frac{1}{2}(\lim_{x \rightarrow x_0^+} f(x) + \lim_{x \rightarrow x_0^-} f(x))$;
- 4 at the ends $f(a) = f(b) = \frac{1}{2}(\lim_{x \rightarrow a^+} f(x) + \lim_{x \rightarrow b^-} f(x))$.

Imprecisely we may say, that if f has a finite number of finite jumps over some bounded interval, then it can be at any point approximated by its Fourier series.